

Resource-NMCTS: Neural Monte Carlo Tree Search with Affine Boolean Preconditioning for Resource-Constrained Quantum Oracle Synthesis

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Abstract

Quantum Boolean oracle synthesis is a central subroutine in many quantum algorithms, yet direct algebraic-normal-form constructions can be expensive in T-count, depth, and ancillary workspace. We study oracle synthesis as a logic-level search problem over compute/uncompute factorizations of Boolean monomials. The core Affine-NMCTS solver combines resource-weighted neural Monte Carlo tree search with affine Boolean input preconditioning, allowing the search to change the XOR coordinate system before factoring the resulting fixed-polarity Reed–Muller representation. The full Resource-NMCTS method adds a budgeted resource-adaptive portfolio guard that selects, under the current weighted objective, among direct ANF, FPRM/affine neural MCTS, and tractable fixed-coordinate MCTS or ESOP cube-beam candidates; it does not use SSHR candidates. On the current evidence-affine benchmark of 322 Boolean functions, Affine-NMCTS is correct on all functions and reduces mean T-count by 61.83% relative to direct ANF synthesis. On a separate $n \leq 6$ traditional-baseline slice, the Pareto-Resource-NMCTS archive has the lowest mean T-count and weighted score, and improves on the strengthened Resource-NMCTS portfolio with 68 score wins, no losses, and 109 ties. It has 174 score wins and no losses against ESOP cube beam search, and 167 score wins, 3 losses, and 7 ties against time-limited weighted ESOP MILP; aggregated by variable count, it reduces total T-count and CNOT count versus ESOP-MILP in every $n = 3, \dots, 6$ group and lowers aggregate peak ancilla for all groups except the tied $n = 3$ case. Against the exported ABC-ESOP baseline on the same 177 functions, it reduces aggregate T-count by 25.80%, CNOT count by 5.40%, peak ancilla by 11.58%, and weighted score by 23.42%, with 170 score wins, 1 loss, and 6 ties. On an exported external comparison, Resource-NMCTS has 69 score wins and 3 losses against the exact $n \leq 4$ CNOT-optimized SSHR-I pilot, 168 score wins and 9 losses against the time-limited $n \leq 6$ SSHR-I extension, and 177 score wins with no losses against an ABC-AIG baseline on the same $n \leq 6$ export. It also has 177 score wins with no losses against ABC-XAG/GIA, ABC-LUT, and BDD/Shannon baselines, and 147 score wins, 24 losses, and 6 ties against ABC’s quantum-cost ESOP minimization path. On a separate exact bounded FPRM-DP slice over the same $n \leq 4$ functions, Resource-NMCTS has 51 score wins, 3 losses, and 18 ties, while Pareto-Resource-NMCTS has 51 wins, no losses, and 21 ties against the restricted exact optimum. An exact XAG multiplicative-complexity slice further shows that Resource-NMCTS reaches the global logical-AND T-count lower bound on 12/72 functions, with a +53.01% mean T gap versus +120.14% for ESOP-MILP and +143.06% for T-optimized SSHR-I. A resource-profile stress test shows that Pareto-Resource-NMCTS is best or tied on 44/47, 44/47, 42/47, and 43/47 functions under T-heavy, balanced, CNOT-depth-heavy, and ancilla-tight objectives, improving on Profile-Resource-NMCTS with no score losses in any profile. A larger core benchmark through $n = 12$ contains 330 functions and 1980 method/function rows; Resource-NMCTS and Profile-Resource-NMCTS complete all functions, and Resource-NMCTS reduces mean T-count by 60.37% relative to direct ANF and by 37.25% relative to logical-AND direct ANF, while fixed-coordinate MCTS

times out on five $n = 12$ rows. An isolated $n = 14$ stress benchmark adds 64 random ANF functions and 576 rows with no errors or skips; the Pareto high-dimensional variant reduces mean T-count by 57.94% relative to direct ANF and by 33.30% relative to logical-AND direct ANF, with 60 score wins, no losses, and 4 ties against FPRM-greedy. A follow-up $n = 15$ scale check preserves this pattern with 30 score wins, no losses, and 2 ties against FPRM-greedy, reducing mean weighted score by 5.73–5.75%. A further $n = 16$ scale check contains 24 random ANF functions and 168 rows with no errors or skips; the Pareto archive reduces mean T-count by 62.12% relative to direct ANF and has 22 score wins, no losses, and 2 ties against bounded FPRM root-beam. On a final $n = 18$ stress check, 12 random ANF functions produce 84 verified rows with no errors or skips; a fast linear-pair guard improves bounded FPRM root-beam by 1.91% in mean weighted score, and the guarded Resource/Profile/Pareto variants reduce mean T-count by 60.05% and mean weighted score by 56.99% relative to direct ANF, with 12 score wins and no losses against root-beam. On exported high-dimensional ABC AIG/XAG/LUT and BDD checks for the same $n = 14$, $n = 15$, $n = 16$, and $n = 18$ sets, the guarded variants win all 528 weighted-score comparisons with 93.24%–99.69% mean score reductions. At $n = 18$ the mean score reductions are 98.85% against ABC-AIG, 99.05% against ABC-XAG, 99.69% against ABC-LUT, and 98.30% against BDD. ABC-AIG and ABC-XAG keep lower estimated depth on most functions, while ABC-LUT and BDD are deeper under our sequential network estimates. The method is evaluated only at the logical circuit layer; hardware mapping and connectivity constraints are outside the present scope.

1 Introduction

Boolean oracles translate classical predicates into reversible quantum subroutines. In fault-tolerant settings, the cost of such oracles is often dominated by non-Clifford resources, so synthesis methods must account for T-count, depth, and ancillary workspace rather than only the number of reversible gates. Existing Boolean-oriented approaches provide strong representations and baseline synthesis methods, while recent learning-guided circuit work shows that neural policies and tree search can be useful in large synthesis spaces. The gap addressed here is the combination of these two lines: a neural search method whose state and actions remain symbolic Boolean oracle constructions and whose objective directly scores T-count, CNOT count, depth, gates, and ancilla.

This paper frames quantum Boolean oracle synthesis as a resource-constrained search over algebraic-normal-form (ANF) and fixed-polarity Reed–Muller (FPRM) factorizations. The central claim is that affine changes of Boolean coordinates can expose low-T decompositions that fixed-coordinate ANF factoring misses, and that a resource-adaptive neural MCTS portfolio can retain these wins while guarding against degradation under different weighted objectives.

2 Related Work and Gap

Boolean-structure methods are the closest technical baselines. XAG-based quantum compilation uses XOR-and-AND logic networks to reduce fault-tolerant resources; ROS combines LUT mapping with resource-aware oracle synthesis and SAT-based garbage management; multiplicative-complexity methods relate the number of AND nodes in a Boolean network to T-count and ancilla upper bounds; back-end-aware oracle synthesis extends XAG-oriented optimization toward backend-sensitive T-count, logical-time-step, and helper-qubit objectives; ABC provides a mature AIG rewriting and verification baseline for Boolean logic; and BDD-based reversible synthesis scales to large Boolean functions by compiling decision-diagram structure [5, 6, 4, 11, 1, 10]. SSHR is also closely related because it exploits parallelotope structure in the Boolean hypercube and provides CNOT-oriented small-function baselines, including heuristic and ILP variants [12]. These methods motivate the

resource metrics and symbolic action space used here, but they do not use a learned policy/value search over ANF/FPRM factorization plans.

Learning-guided synthesis has developed along adjacent axes. ShortCircuit uses an AlphaZero-style process for classical AIG generation from truth tables, showing that policy/value search can help Boolean circuit construction at the logic level [9]. Quantum learning-guided work has used Gumbel AlphaZero for Clifford+T unitary synthesis, reinforcement learning for non-Clifford-count optimization, MCTS for Hamiltonian-simulation Pauli-term ordering, and PPO/GNN policies for ZX-calculus rewriting [7, 2, 3, 8]. These systems validate the use of neural search in synthesis, but they do not directly target Boolean oracle factorization with explicit T/CNOT/depth/ancilla tradeoffs.

The present method therefore avoids two weak directions for this problem: it is not a gate-by-gate unitary generator, and it is not an SSHR parallelotope selector. It searches over reusable Boolean intermediates in ANF/FPRM form, uses affine coordinate changes as a cheap way to alter the factorization basis, and treats external XAG, ROS, BDD, ESOP, and SSHR-style flows as baselines or export targets rather than as components of the method.

3 Problem Formulation

Let $f : \{0, 1\}^n \rightarrow \{0, 1\}$ be a Boolean function. The target oracle maps

$$|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle.$$

We represent f by its ANF,

$$f(x) = \bigoplus_{m \in M} \prod_{i \in m} x_i,$$

where each monomial is encoded as a bit mask. Direct ANF synthesis applies one controlled operation per monomial. The search problem is to replace this direct construction by reusable compute/uncompute subplans that lower the weighted resource score

$$S = w_T T + w_C CNOT + w_D D + w_G G + w_A A,$$

where T is T-count, D is logical depth, G is gate count, and A is peak ancilla.

4 Method

4.1 Resource-aware ANF factorization

At each search state, the remaining ANF terms are a set of monomial masks. An action chooses a common factor S appearing in at least two terms, computes S into a clean temporary ancilla, synthesizes the residual terms using that ancilla as an additional control, uncomputes S , and then synthesizes the remaining terms. The implementation restricts the number of live factor ancillae and ranks candidate factors by immediate weighted resource gain.

4.2 Logical-AND cost model

The reported resource objective uses a logical-AND accounting mode: temporary ANDs are charged four T gates for compute and T-free Clifford/measurement-based cleanup for uncompute. The verifier circuit remains deterministic and is checked by exhaustive classical simulation over the truth table. Thus the paper reports logic-level resource estimates rather than mapped hardware circuits.

4.3 Neural Monte Carlo tree search

The MCTS solver recursively applies PUCT-style selection to factorization actions. Unvisited subproblems use a greedy rollout value. A small neural action scorer is trained from synthetic ANF states with rollout-improvement labels, then used as an additive prior during action ranking.

4.4 Affine Boolean preconditioning

Affine-NMCTS searches invertible linear transforms $y = Bx$ before ANF/FPRM factorization. For each candidate transform, it synthesizes $h(y) = f(B^{-1}y)$ and wraps the body circuit by CNOT networks implementing and undoing B . Candidate transforms include the identity, star-like XOR transforms, single-CNOT transforms, and random short CNOT sequences. The candidate list is first ranked by fast FPRM/greedy factoring; low-dimensional instances are then refined by neural MCTS, while large instances use term-count guards to avoid affine densification.

4.5 Guard against fixed-coordinate degradation

The affine plan is compared against a fixed-coordinate MCTS plan under the same weighted score. The lower-score plan is returned. This guard makes the method nondegrading in score on completed fixed-coordinate MCTS pairs, while preserving affine improvements when coordinate changes reveal better factorizations.

4.6 Resource-adaptive portfolio guard

The full Resource-NMCTS solver treats the weighted resource objective as a selection criterion over internally generated candidates. For each function it evaluates direct ANF, FPRM/greedy factoring, and a budgeted Affine-NMCTS candidate. When the configured dimension guard permits it, the portfolio also evaluates fixed-coordinate logical-AND MCTS; for $n \leq 6$, it additionally evaluates ESOP cube-beam search. For $n > 12$, where full fixed-coordinate MCTS and broad affine search create long tails, it replaces the expensive branches by a bounded recursive FPRM linear-pair candidate: many polarities are screened by direct cost, a root-child beam baseline is retained, and repeated term pairs $x_i m \oplus x_j m$ can be factored as $(x_i \oplus x_j)m$ using a temporary CNOT-computed linear control. One extra recursive layer applies the same pair search to both the quotient and the remaining terms when the subproblem has at most 900 terms. It returns the correct candidate with the smallest weighted score, breaking ties by T-count, CNOT count, depth, and peak ancilla. SSHR is used only as an external baseline in the experiments and is not part of this portfolio. This final guard converts the affine neural search into a resource-constrained synthesis method: when affine coordinates are best, they are retained; when a simpler internal construction is better under the active weights, the method can select it. The guard is budgeted rather than an all-scale dominance certificate: high-dimensional runs deliberately avoid internal candidates that create long runtime tails.

4.7 Profile-aware candidate generation

Profile-Resource-NMCTS adds a second portfolio layer for explicit resource profiles. For $n \leq 6$, it evaluates additional T-aggressive, CNOT/depth-oriented, and ancilla-tight FPRM, affine-greedy, and ESOP cube-beam candidates before selecting by the active weighted score. For larger random-ANF instances, the same method delegates to the high-dimensional Resource-NMCTS guard: it keeps direct ANF and the bounded recursive FPRM linear-pair candidate, but avoids expensive

fixed-coordinate MCTS guards. A width-three linear-parity factor is reported as a standalone ablation in the $n = 15$ scaling check, but the portfolio uses the recursive pair candidate because it dominates the parity ablation under the weighted objective. This design separates two roles. On small functions it tests whether the resource profile can change the selected circuit; on large functions it acts as a runtime-tail guard rather than a score-dominance guarantee.

4.8 Pareto resource archive

Pareto-Resource-NMCTS strengthens the small-function resource-profile search by changing the internal objective used to generate candidates, not only the search budget. For each function it evaluates child searches under five candidate-generation objectives: the active score, a T-sparse score, a CNOT/depth-heavy score, an ancilla-tight score, and a gate-count-sensitive score. The resulting candidates include FPRM greedy, bounded root-child beam, CNOT-only linear-pair factoring, affine-greedy, selected affine-neural, and ESOP cube-beam plans when the dimension guard permits them. The method then removes candidates dominated simultaneously in T-count, CNOT count, depth, gate count, and peak ancilla, and finally selects from this non-dominated archive using the active resource profile. Thus the method keeps the same logical verifier and cost model as Resource-NMCTS, but explicitly approximates a local Pareto frontier before profile-specific selection. On $n \leq 6$ functions, the archive also includes a ranked polarity candidate: all 2^n FPRM polarities are screened under greedy, root-child-beam, and CNOT-only linear-pair factorizations, and the best scored plan is added to the non-dominated pool.

5 Experiments

5.1 Benchmark and baselines

The current evidence.affine run contains 322 Boolean functions: structured oracles such as parity, majority, threshold, mux, adders, and multipliers; random truth tables for $n = 4, 5, 6$; and random ANF functions for $n = 8, 10, 12$. We compare: direct ANF, logical-AND direct ANF, fixed-coordinate logical-AND MCTS, Affine-NMCTS, and SSHR-H for functions with $n \leq 6$. A separate traditional_resource slice restricts the benchmark to $n \leq 6$ and adds Resource-NMCTS, ESOP cube beam search, and a time-limited weighted ESOP MILP over candidate cubes. The large_resource_core benchmark extends the same resource objective to a larger random-ANF core through $n = 12$. To support external reproduction, the benchmark generator can export every preset to PLA, BLIF .names, and truth-table JSON manifests. The current draft adds an exported exact-SSHR-I pilot on the $n \leq 4$ subset, a time-limited $n \leq 6$ SSHR-I extension, Berkeley ABC AIG, XAG/GIA, LUT, and ESOP baselines, and a reduced ordered BDD baseline covering the full traditional_resource slice. The ABC-AIG, ABC-XAG, ABC-LUT, and BDD export paths are also run on the isolated $n = 14$, $n = 15$, and $n = 16$ high-dimensional random-ANF suites; reproduced ROS/mockturtle reversible-toolchain results are still future work.

5.2 Main results

Affine-NMCTS was correct on all 322 functions. Fixed-coordinate MCTS timed out on one $n = 12$ random ANF function after a 600 s repair attempt; Affine-NMCTS completed the same function within the experiment budget and produced a valid circuit. The largest direct-ANF reductions exceeded 80% T-count on several structured and random small functions.

Table 1: Current evidence.affine results. Improvements are relative T-count means over functions where both methods completed.

Comparison	Functions	Wins	Ties	Mean ΔT
Affine-NMCTS vs direct ANF	322	286	36	−61.83%
Affine-NMCTS vs logical-AND direct ANF	322	284	38	−40.12%
Affine-NMCTS vs fixed-coordinate MCTS	321	161	160	−16.18%
Affine-NMCTS vs SSHR-H	177	171	5	−43.89%

Table 2: Mean logical resources on the traditional_resource slice. ESOP MILP is a time-limited weighted ESOP optimization over candidate cubes.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	184.1	134.2	134.6	1.33	194.3
AND-direct ANF	101.0	166.5	166.9	2.18	115.2
Fixed MCTS	62.1	124.3	124.8	1.82	73.1
Affine-NMCTS	45.9	94.4	98.9	1.88	55.4
Resource-NMCTS	43.9	89.9	94.5	1.92	53.2
Polarity archive	43.0	86.9	91.5	2.11	52.5
Pareto-Resource-NMCTS	40.4	83.0	88.0	2.03	49.6
ESOP cube beam	71.3	115.3	139.5	2.55	83.8
ESOP MILP	83.6	133.5	159.6	2.32	96.7
SSHR-H	81.0	67.6	88.7	1.36	88.2

5.3 Traditional ESOP baselines

The traditional_resource slice tests whether the gain persists against traditional Boolean/ESOP-style synthesis rather than only direct ANF and MCTS variants. It contains 177 functions with $n \leq 6$ and ten methods, producing 1770 method/function rows with no errors or skips. On this slice, Pareto-Resource-NMCTS had the lowest mean T-count and the lowest mean composite score among all tested methods.

The ESOP-focused aggregate in Table 3 is the same-benchmark comparison corresponding most closely to the ESOP column used in prior small-function studies. It should not be read as a direct numerical comparison to published ESOP totals from a different function set or cost model.

Against the strengthened Resource-NMCTS, Pareto-Resource-NMCTS had 68 score wins, no score losses, and 109 ties, with a 3.26% mean composite-score reduction. Against ESOP cube beam search, Pareto-Resource-NMCTS had 174 score wins, no losses, and 3 ties, with a 35.86% mean composite-score reduction. Against weighted ESOP MILP, Pareto-Resource-NMCTS had 167 score wins, 3 losses, and 7 ties, with a 29.84% mean composite-score reduction. Against SSHR-H, Pareto-Resource-NMCTS had 173 T-count wins, no T-count losses, 4 ties, and 173 score wins with 4 score losses. SSHR-H retained the lowest mean CNOT count on this small-function slice, so this result supports low-T and weighted-resource synthesis rather than CNOT-only optimality. The aggregate ESOP comparison is stronger than the per-function win/loss counts alone: against internal ESOP-MILP, Pareto-Resource-NMCTS lowers total T-count by 51.64%, CNOT count by 37.80%, and summed peak ancilla by 12.65%. Against external ABC-ESOP, it lowers total T-count by 25.80%, CNOT count by 5.40%, and summed peak ancilla by 11.58%. After adding the cheap no-MCTS linear/root-beam candidates to the Resource portfolio, Pareto-Resource-NMCTS has lower aggregate T-count, CNOT count, peak ancilla, and weighted score in every $n = 3, \dots, 6$

Table 3: ESOP-centered aggregate comparison by variable count on the matched $n \leq 6$ benchmark. T/C/A denotes total T-count, total CNOT count, and summed peak-ancilla count across functions in the group. Negative changes favor Pareto-Resource-NMCTS.

n	ESOP baseline	Func.	ESOP T/C/A	Ours T/C/A	ΔT	Δ CNOT	ΔA	Score W/L/T
3	Internal ESOP-MILP	3	24/45/0	8/27/0	-66.7%	-40.0%	+0.0%	2/0/1
4	Internal ESOP-MILP	69	1456/2466/102	1028/2217/100	-29.4%	-10.1%	-2.0%	64/1/4
5	Internal ESOP-MILP	67	3392/5558/169	2580/5461/139	-23.9%	-1.7%	-17.8%	65/1/1
6	Internal ESOP-MILP	38	9924/15562/140	3540/6993/120	-64.3%	-55.1%	-14.3%	36/1/1
3	ABC-ESOP export	3	24/45/0	8/27/0	-66.7%	-40.0%	+0.0%	2/0/1
4	ABC-ESOP export	69	1576/2622/104	1028/2217/100	-34.8%	-15.4%	-3.8%	66/0/3
5	ABC-ESOP export	67	3592/5802/170	2580/5461/139	-28.2%	-5.9%	-18.2%	66/0/1
6	ABC-ESOP export	38	4452/7068/132	3540/6993/120	-20.5%	-1.1%	-9.1%	36/1/1

ABC-ESOP group.

5.4 Exported external baselines

To check the external-tool exchange path, we exported the traditional_resource benchmark to PLA, BLIF, and truth-table JSON, then ran separate external backends over the exported manifest. The SSHR backend gives CNOT-oriented heuristic and ILP baselines. The ABC-AIG backend optimizes the exported BLIF with ABC rewriting, verifies the optimized BLIF truth table, and maps AIG AND count and level to a logic-level Bennett compute/uncompute resource estimate. The ABC-XAG backend uses ABC’s `GIA &st -m -L 1` path, records XOR/MUX/AND statistics with `&ps -m -x`, verifies the emitted BLIF truth table, and scores the network with a Bennett-style logical XAG estimate. The ABC-ESOP backend runs ABC’s quantum-cost `&exorcism -q` flow, verifies the emitted ESOP-PLA against the exported truth table, and scores the resulting XOR-of-cubes oracle using the same logical-AND cube model as the internal ESOP baselines. The ABC-LUT backend maps the same BLIF through `if -K 4`, verifies the mapped BLIF, and scores a compute/output/uncompute network whose local LUT functions are converted to ANF terms. The BDD backend builds reduced ordered BDDs under several deterministic variable orders, verifies the decision diagram against the exported truth table, and scores a compute/uncompute Shannon network. For the 72 functions with $n \leq 4$, the SSHR-I runs are exact within the configured Gurobi solve. This exact pilot produced 216 external rows with no errors or skips and confirmed that Resource-NMCTS retains a weighted-score advantage against both CNOT-optimized and T-optimized SSHR-I, while spending more CNOTs.

The exact $n \leq 4$ pilot strengthens but also sharpens the claim. Against CNOT-optimized SSHR-I, Resource-NMCTS has 65 T-count wins, no T-count losses, and 7 ties; by weighted score it has 69 wins, 3 losses, and no ties, with a 26.45% mean score reduction. Against T-optimized SSHR-I it shows the same T-count win/tie pattern and a 26.21% mean score reduction. In both exact comparisons, however, Resource-NMCTS loses CNOT count on most functions. This supports the resource-constrained low-T oracle-synthesis claim, while ruling out any CNOT-only optimality claim.

Table 5 adds an internal exact slice for the Boolean factorization model. The Exact FPRM-DP solver enumerates all polarities and uses dynamic programming over all monomial factors and CNOT-only linear-pair factors under the same logical-AND resource model. It completed all 72 $n \leq 4$ functions with no errors or skips. Resource-NMCTS improves on this restricted exact optimum by 51/3/18 score wins/losses/ties and a 12.18% mean score reduction, while Pareto-Resource-NMCTS improves it by 51/0/21 and 12.20%. Because the DP is exact only inside this bounded FPRM factor family, these wins mean the portfolio found candidates outside that restricted model. The exact

DP baseline itself also improves weighted score over ESOP-MILP and exact SSHR-I, but it usually spends more CNOTs than SSHR-I.

To obtain a model-independent T-count reference, we also computed exact XAG multiplicative complexity for the same $n \leq 4$ subset. All 72 functions were solved with no unknown rows. Resource-NMCTS and Pareto-Resource-NMCTS reach the global logical-AND T lower bound on 12 functions and have a +53.01% mean T gap to the bound, whereas ESOP-MILP reaches it on 3 functions with a +120.14% gap and the T-optimized SSHR-I rows reach it on 5 functions with a +143.06% gap. This is not a full reversible-circuit optimum, but it strengthens the claim that the portfolio is substantially closer to the multiplicative-complexity T limit than the ESOP and SSHR-I references on this exact small slice.

We then ran the external extension over all 177 exported functions with $n \leq 6$. This produced 1416 external rows with no errors or skips: 531 SSHR rows, 177 ABC-AIG rows, 177 ABC-XAG rows, 177 ABC-LUT rows, 177 BDD rows, and 177 ABC-ESOP rows. Against the CNOT-optimized SSHR-I extension, Resource-NMCTS had 164 T-count wins, 3 losses, and 10 ties, and 168 score wins with 9 score losses; the mean score reduction was 27.92%. Against the T-optimized extension, Resource-NMCTS had 166 T-count wins, 1 loss, and 10 ties, with a 26.25% mean score reduction. The CNOT disadvantage persisted, with Resource-NMCTS worse on 168/177 CNOT comparisons against the CNOT-optimized run.

The ABC-AIG comparison strengthens the resource-constrained claim from a different direction. Resource-NMCTS had 170 T-count wins, 2 losses, and 5 ties against ABC-AIG; it won all 177 CNOT, peak-ancilla, and weighted-score comparisons, reducing mean weighted score by 54.52% and mean peak ancilla by 86.29%. ABC-AIG had a lower depth estimate on 126/177 functions, which is consistent with a shallow optimized AIG followed by a wide Bennett embedding. This makes ABC-AIG a useful foil for the ancilla and weighted-resource objective, not a claim that Resource-NMCTS dominates every metric.

The ABC-XAG/GIA comparison adds an XOR-aware network baseline without using an external mapper inside the method. Resource-NMCTS had 176 T-count wins, no losses, and 1 tie against ABC-XAG; it won all 177 peak-ancilla and weighted-score comparisons, reducing mean weighted score by 63.23% and mean peak ancilla by 89.85%. It also reduced mean CNOT by 35.53%, although ABC-XAG retained a lower depth estimate on 144/177 functions. This closes part of the earlier XAG comparison gap while still falling short of a full ROS or mockturtle reversible-toolchain reproduction.

The ABC-LUT comparison adds a mapped-LUT network baseline closer to ROS/LUT families, though still without a full reversible LUT mapper. Resource-NMCTS won all 177 CNOT, depth, peak-ancilla, and weighted-score comparisons against ABC-LUT, with 175 T-count wins and 2 ties. The mean weighted-score reduction was 76.41%. This result shows that a naive compute/uncompute embedding of LUT-mapped BLIF is not competitive under the current logical-resource model, but it narrows the external LUT comparison gap.

The BDD comparison adds a scalable decision-diagram baseline. Resource-NMCTS won all 177 T-count, CNOT, depth, peak-ancilla, and weighted-score comparisons against BDD, reducing mean weighted score by 67.15%. This result should be read as a comparison to a verified logic-level Shannon-network embedding of ROBDDs rather than to every optimized BDD-based reversible synthesis variant.

The ABC-ESOP comparison is closer to the XOR-of-products synthesis family. Resource-NMCTS had 144 T-count wins, 19 losses, and 14 ties against ABC-ESOP, and 147 score wins, 24 losses, and 6 ties. The mean weighted-score reduction was 19.88%; unlike the SSHR-I CNOT-oriented comparison, Resource-NMCTS also had a small mean CNOT reduction of 1.19%. This result is not a substitute for a reproduced ROS mapper, but it adds a standard external XOR/ESOP minimizer to the comparison boundary. The stronger small-function Pareto-Resource-NMCTS setting improves

this ESOP boundary: on the same ABC-ESOP rows it has 170 score wins, 1 score loss, and 6 ties, with a 23.42% aggregate score reduction. Its aggregate T-count and peak-ancilla sums are lower in every $n = 3, \dots, 6$ group; aggregate CNOT is lower through $n = 5$ and nearly tied at $n = 6$.

Finally, we pushed the ABC AIG/XAG/LUT and BDD exchange beyond the small-function slice. The `highdim_resource`, `highdim_scale_resource`, `ultra_highdim_resource`, and `mega_highdim_resource` suites were exported to BLIF/truth JSON, optimized or represented by the external-style baselines, and verified by a bit-parallel full truth-table checker before scoring. The high-dimensional rows contain 64 $n = 14$, 32 $n = 15$, 24 $n = 16$, and 12 $n = 18$ functions for each of ABC-AIG, ABC-XAG, ABC-LUT, and BDD, all correct and with no errors or skips. Against all four baselines, Resource-NMCTS, Profile-Resource-NMCTS, and Pareto-Resource-NMCTS win every T-count, CNOT, peak-ancilla, and weighted-score comparison across these 528 external rows; the mean score reductions range from 93.24% to 99.69%. At $n = 18$, the weighted-score reductions are 98.85% against ABC-AIG, 99.05% against ABC-XAG, 99.69% against ABC-LUT, and 98.30% against BDD. The depth column remains an important counterweight for ABC: ABC-AIG has a lower estimated depth on 50/64, 25/32, 22/24, and 10/12 functions across the four high-dimensional suites, and ABC-XAG has a lower estimated depth on 50/64, 26/32, 22/24, and 11/12 functions, respectively. ABC-LUT and BDD depths are much higher under the sequential network estimates. Thus the high-dimensional external result supports the low-ancilla and weighted-resource claim, but not depth-only dominance against optimized ABC networks.

5.5 Ablation

The `ablation_affine` run used the same 322-function suite and added two variants: `affine-greedy`, which performs affine transform search followed by FPRM/greedy factoring, and `affine-no-guard`, which adds neural refinement but does not compare against fixed-coordinate MCTS. This ablation separates the effect of coordinate search, neural refinement, and the final guard.

Affine-greedy alone reduced mean T-count by 60.92% relative to direct ANF, showing that the affine coordinate search is the dominant contributor. Neural refinement and the guard were smaller but monotone under the composite score: neither introduced score losses on the benchmark.

We also reran the $n \leq 6$ `traditional_resource` slice with the learned action scorer removed while keeping the same heuristic PUCT/action prior. The paired comparison contains 1062 rows over Affine-NMCTS, Resource-NMCTS, and Pareto-Resource-NMCTS, with no errors or skips. The learned prior gives modest but monotone score improvements: 42/0/135 score wins/losses/ties for Affine-NMCTS, 39/0/138 for Resource-NMCTS, and 29/0/148 for Pareto-Resource-NMCTS. Mean score reductions are 1.47%, 1.10%, and 0.78%, respectively. The cost is runtime: the paired mean relative runtime increases are 91.22%, 55.11%, and 18.77%, because many small neural action batches are evaluated on CPU. We therefore treat the learned prior as a quality-improving search signal, not as the current fastest mode. For the high-dimensional Resource-NMCTS path, we also tested a dedicated linear-action scorer with root-only neural screening on 12 $n = 14$ random ANF functions. This diagnostic produced only one score win and eleven ties (-0.01% mean score change) while roughly doubling mean runtime, so we do not claim a strong high-dimensional learned-prior effect in the current version. To separate “no headroom” from “weak labels”, we additionally evaluated a bounded root-action teacher for the same high-dimensional slice. The teacher scores a wider CNOT-only linear-pair root-action set by building actual greedy child plans. As Table 12 shows, oracle top-24 selection has 3/0/7 score wins/losses/ties over the current heuristic top-4 window and 7/0/3 over heuristic top-1. The effect is small, but positive. The current neural top-4 ordering does not exploit it, scoring 1/1/8 against the heuristic top-4 window. We therefore treat this as a supervised target for a future high-dimensional root-action ranker rather than as a present

headline improvement.

Table 13 consolidates the scattered ablation evidence into one matched-pair contribution account. The strongest pre-portfolio algorithmic step is affine coordinate search: affine-greedy already reduces the composite score by 12.12% relative to completed fixed-coordinate MCTS pairs, albeit with score losses on some functions. The neural refinement and final guard are smaller but monotone in score on this suite, and the learned prior gives a positive no-loss quality signal on the $n \leq 6$ rerun; the separate $n = 14$ learned-prior diagnostic is kept as a boundary result rather than a main contribution. The root-action teacher diagnostic shows a small learnable high-dimensional ranking signal, but the current immediate-label scorer does not yet capture it. A dedicated small-function rerun further separates heuristic-only, beam-only, no-MCTS, Resource-NMCTS, and Pareto-Resource-NMCTS portfolios on the same 177 functions. The table also clarifies the high-dimensional claim. At $n = 14$ – $n = 18$, most of the measurable scale gain comes from bounded linear-pair guards and their recursive/fast variants. A lightweight $n = 14$ high-dimensional ablation confirms that the no-MCTS guard portfolio improves both root-beam and linear-pair children on 14 functions with no score losses among 16 matched random ANF functions. Pareto or Resource selection provides additional separation in some suites, but should not be described as a large independent archive effect when it ties the guarded child.

5.6 Runtime and resource tradeoff

The ablation also exposes the computational cost of the search components. Affine-greedy completed all 322 functions with a 0.03 s median runtime and a 1.82 s maximum runtime, while the full Affine-NMCTS method had a 0.61 s median runtime and a 300.02 s maximum runtime because it invokes the fixed-coordinate MCTS guard. Fixed-coordinate MCTS timed out on one $n = 12$ function, whereas all affine variants completed every function.

5.7 Resource-profile stress test

To test whether the method remains competitive under different resource objectives, we ran a 47-function $n \leq 6$ sweep under four profiles: T-heavy, balanced, CNOT-depth-heavy, and ancilla-tight. The sweep produced 1692 method/profile/function rows with no errors or skips. Scores are comparable within a profile but not across profiles because each profile uses a different weighted objective.

Pareto-Resource-NMCTS was best or tied on 44/47 functions under T-heavy weights, 44/47 under balanced weights, 42/47 under CNOT-depth-heavy weights, and 43/47 under ancilla-tight weights. Relative to Profile-Resource-NMCTS, it had score wins/losses/ties of 19/0/28, 19/0/28, 21/0/26, and 15/0/32 across the four profiles, with mean score reductions of 5.56%, 4.93%, 4.85%, and 2.51%. Relative to Resource-NMCTS, it also had no score losses in any profile. The mean Pareto-Resource-NMCTS resource vector changed with the active objective: mean T-count stayed at 35.91 for T-heavy, balanced, and CNOT-depth profiles and rose to 39.74 under ancilla-tight weights, while mean peak ancilla fell from 1.87–1.94 to 1.62 in the ancilla-tight profile. This result shows that objective-specific candidate generation and non-dominated archive selection produce a stronger profile-sensitive Pareto search than the earlier bounded profile portfolio.

5.8 Large-scale core benchmark

The `large_resource_core` run contains 330 functions, including structured oracles, random truth-table functions up to $n = 6$, and random ANF functions up to $n = 12$. It produces 1980 method/function rows under process-isolated hard timeouts on an Apple M4 Pro workstation with 14 logical CPU

cores and 24 GB memory. Direct ANF, logical-AND direct ANF, Affine-NMCTS, Resource-NMCTS, and Profile-Resource-NMCTS complete all 330 functions; fixed-coordinate MCTS completes 325 functions and times out on five $n = 12$ random ANF rows.

On this larger suite, Resource-NMCTS reduces mean T-count by 60.37% versus direct ANF and by 37.25% versus logical-AND direct ANF. The corresponding composite-score reductions are 56.84% and 35.21%. Against completed fixed-coordinate MCTS pairs, Resource-NMCTS has 139 T-count wins, 15 losses, and 171 ties, with an 11.41% mean composite-score reduction. This comparison should be interpreted with the five fixed-MCTS timeouts in mind: the completed-row mean for fixed MCTS excludes its hardest rows, whereas Resource-NMCTS completed all functions. Relative to standalone Affine-NMCTS, the budgeted portfolio has 29 score wins, 16 score losses, and 285 ties, with a 0.20% mean score reduction. Thus the portfolio is useful as a scalable resource-aware guard, but it is not a dominance guarantee over the full affine search when high-dimensional instances force tighter candidate budgets. Profile-Resource-NMCTS has a lower runtime tail than Resource-NMCTS on this suite, with p95 runtime 25.68 s versus 58.86 s, but its cheap high-dimensional guard increases mean score from 273.72 to 286.54. We therefore treat it as a profile-sensitive and tail-reducing extension, not as the large-scale score winner.

5.9 High-dimensional $n=14$ stress test

The `highdim_resource` run isolates high-dimensional random ANF instances rather than mixing them with small structured functions. It contains 64 $n = 14$ functions and nine methods, for 576 method/function rows with no errors or skips. This experiment uses the high-dimensional guard described above: fixed-coordinate MCTS and broad affine search are avoided inside the guarded methods, while direct ANF, bounded FPRM root-child beam, linear-pair candidates, and bounded Pareto archive variants remain available.

The Resource/Profile/Pareto guarded methods complete all 64 functions. Relative to direct ANF, Pareto-Resource-NMCTS has 61 T-count wins, no losses, and 3 ties, with a 57.94% mean T-count reduction and a 54.55% mean composite-score reduction; Resource-NMCTS and Profile-Resource-NMCTS have the same win/tie pattern with 57.43% and 54.01% reductions. Relative to logical-AND direct ANF, Pareto-Resource-NMCTS again has 61 score wins, no losses, and 3 ties, with a 31.40% mean score reduction. Against FPRM-greedy, the guarded methods have 60 weighted-score wins, no losses, and 4 ties, with 6.29–6.83% mean score reductions. Against the standalone FPRM root-beam candidate, Pareto-Resource-NMCTS has 60 score wins, no losses, and 4 ties, with a 6.31% mean score reduction. Against the active Resource/Profile candidates, Pareto-Resource-NMCTS gives 16 score wins, no losses, and 48 ties, reducing mean score by 0.59%. The mean T-count falls from 2263.69 for FPRM-greedy, 2248.81 for root beam, and 2193.50 for one-layer linear pair to 2149.50 for Resource-NMCTS and 2148.81 for Pareto-Resource-NMCTS. The improvement is mechanistically different from simply widening the root beam: the recursive linear-pair candidate computes $x_i \oplus x_j$ into a temporary ancilla with CNOTs, factors the paired residual subplan under that control, and then recursively searches the quotient and bounded rest subproblems before uncomputing the linear control. Runtime tails remain bounded but increase: Resource-NMCTS has median runtime 8.57 s and p95 runtime 80.43 s, Profile-Resource-NMCTS has median runtime 9.29 s and p95 runtime 84.23 s, while Pareto-Resource-NMCTS has median runtime 67.79 s and p95 runtime 300.02 s. The resource tradeoff is workspace: mean peak ancilla increases from 2.03 for FPRM-greedy and root beam to about 3.05–3.06 for the guarded variants.

As a scale check, we also run the same high-dimensional guard on 32 random $n = 15$ ANF functions, producing 320 method/function rows with no errors or skips after adding the bounded recursive linear-pair candidate, a standalone width-three linear-parity ablation, and a bounded

Pareto archive. Relative to FPRM-greedy, the Resource/Profile/Pareto guarded variants have 30 T-count wins, no losses, and 2 ties, with 6.89–6.93% mean T-count reductions and 5.73–5.75% mean score reductions. Relative to the standalone FPRM root-beam candidate, they have 30 score wins, no losses, and 2 ties, with 5.28–5.31% mean score reductions. The recursive pair candidate is a material refinement of the one-layer linear-pair search: it improves weighted score in 29 cases, ties in 3, and has no score losses, while increasing mean peak ancilla from 2.84 to 3.06. The Pareto archive extracts a smaller residual gain over the active Resource/Profile candidates, with 5 score wins, no losses, 27 ties, and a 0.03% mean score reduction. Runtime remains finite under the 900-term recursion guard: Resource-NMCTS has median runtime 9.70 s and p95 runtime 90.02 s, Profile-Resource-NMCTS has median runtime 11.33 s and p95 runtime 93.68 s, and Pareto-Resource-NMCTS has median runtime 117.52 s and p95 runtime 300.30 s.

The additional `ultra_highdim_resource` run checks whether the same guard remains usable one dimension higher. It contains 24 random ANF functions at $n = 16$, seven methods, and 168 method/function rows with no errors or skips. For this dimension the ultra-scale Resource/Profile guard deliberately disables the recursive rest-branch linear-pair refinement used at $n = 15$ and keeps the one-layer FPRM linear-pair candidate. Consequently Resource-NMCTS and Profile-Resource-NMCTS match the standalone linear-pair candidate on all 24 functions, while Pareto-Resource-NMCTS adds bounded profile variants around the same high-dimensional candidate set. Relative to direct ANF, Pareto-Resource-NMCTS has 23 T-count wins, no losses, and 1 tie, reducing mean T-count by 62.12% and mean weighted score by 59.56%. Relative to logical-AND direct ANF, it has 23 score wins, no losses, and 1 tie, with a 31.75% mean score reduction. Relative to FPRM root-beam, it has 22 score wins, no losses, and 2 ties, with a 1.97% mean score reduction. Against the active Resource/Profile candidates, Pareto-Resource-NMCTS gives 9 score wins, no losses, and 15 ties, reducing mean score by 0.09%. The tradeoff remains CNOT/depth against plain direct ANF, where the lower-T factored plans use more linear Clifford work, but they reduce both CNOT and depth relative to logical-AND direct ANF; the Pareto archive also increases median runtime to 130.92 s.

The `mega_highdim_resource` run pushes the same harness two more variables beyond the ultra check. It contains 12 random $n = 18$ ANF functions and seven methods, for 84 method/function rows with no errors or skips. This is presented as a runtime-boundary stress test rather than a new Pareto-archive separation. The standalone fast FPRM linear-pair guard has 6 score wins, no losses, and 6 ties against bounded FPRM root-beam, reducing mean weighted score by 1.91%. The guarded Resource/Profile/Pareto variants then improve that fast child on all 12 functions, reducing mean weighted score by another 3.55%. Relative to direct ANF, these guarded variants have 12 T-count wins, no losses, and no ties, reducing mean T-count by 60.05% and mean weighted score by 56.99%. Relative to bounded FPRM root-beam, they have 12 score wins, no losses, and no ties, with a 5.29% mean score reduction. The price is runtime: median guarded runtime is about 83.3–83.5 s and p95 runtime is about 172.9–173.8 s.

6 Discussion

The results support the central hypothesis that affine Boolean coordinates are a useful search dimension for quantum oracle synthesis and that a resource-adaptive internal portfolio can reduce avoidable score regressions under explicit budget guards. The wins are strongest in T-count and composite resource score. Against SSHR-H, the method often uses more CNOTs and depth, which is expected because SSHR-H is a CNOT-oriented small-function baseline and is not included in the Resource-NMCTS portfolio. The present contribution should therefore be framed as low-T,

resource-constrained logical synthesis, not as a CNOT-only optimizer.

The runtime results show a practical tradeoff. Affine-greedy captures most of the T-count gain at very low runtime, making it the best fast setting in the current implementation. Resource-NMCTS is slower than Affine-NMCTS on the $n \leq 6$ slice because it evaluates additional internal candidates, but its median runtime remains below 1 s in the current small-function experiment. On the large_resource_core benchmark, Resource-NMCTS completes every function with median runtime 1.31 s and p95 runtime 58.86 s; the fixed-coordinate MCTS baseline has five timeouts. The guarded Profile-Resource-NMCTS variant reduces the p95 runtime to 25.68 s on the same large benchmark, at the cost of a higher mean score. On the isolated $n = 14$ stress test, the Resource/Profile/Pareto guarded variants complete all 64 functions and improve over FPRM-greedy with no score losses; the $n = 16$ scale check also completes without errors after switching to the ultra-scale one-layer linear-pair guard and adding a bounded Pareto archive. The $n = 18$ stress check completes all 84 rows after switching to a fast linear-pair high-dimensional guard, and the guarded Resource/Profile/Pareto variants improve bounded root-beam on all 12 functions while keeping minute-scale runtime tails. This shows that the current high-dimensional guard is no longer only a scalability fallback: pairwise XOR factoring can correct a measurable part of the greedy Reed–Muller search error using only Clifford linear preprocessing around the factored subplan.

The main current limitations are the absence of hardware mapping, the fact that the exported SSHR-I extension beyond $n \leq 4$ is time-limited rather than an exact certificate, the fact that ABC/BDD/ESOP/LUT rows are scored through logic-level estimates rather than full reversible-toolchain synthesis, and the lack of a reproduced external ROS/mockturtle toolchain in the comparison set. Accordingly, the results should not be read as a back-end-aware or mapping-level claim. The exact FPRM-DP slice reduces this concern for a small same-model subset, but it is exact only within the bounded fixed-polarity FPRM factor model, not over all reversible circuits. At $n = 16$, the guarded Resource/Profile variants tie their own one-layer linear-pair child, and the Pareto archive improves that child only modestly, so the ultra result is bounded scalability evidence plus a residual archive gain rather than a large new search-quality separation. The same caution applies even more strongly at $n = 18$, where the guard chooses bounded root-beam search and the result is a tractability and verification-boundary check rather than evidence that the full neural portfolio beats its selected child. The in-harness ESOP cube-beam and weighted-MILP baselines plus the exported SSHR-I/ABC/BDD/LUT comparisons strengthen the Boolean synthesis comparison, and the new high-dimensional ABC/BDD/LUT rows show that this external comparison is no longer restricted to $n \leq 6$. The $n = 16$ and $n = 18$ external rows preserve the weighted-score advantage, but they also keep the ABC-AIG/XAG depth caveat. More external implementations remain necessary before a submission-ready claim. The resource-profile sweep shows that objective-specific candidate generation and non-dominated archive selection improve small-function profile adaptation. The large benchmark still shows the corresponding high-dimensional tradeoff: conservative profile guards reduce runtime tails but give up some score. The $n = 14$ stress test narrows this limitation further: bounded recursive FPRM linear-pair search keeps the run finite and improves over FPRM-greedy on most functions, and the $n = 15$ – $n = 16$ scale checks preserve that margin while showing that Pareto archives can extract small residual gains at substantial runtime cost. The remaining weakness is less the internal high-dimensional guard than the comparison boundary: the current evidence still lacks reproduced ROS or mockturtle reversible-toolchain baselines.

7 Conclusion

We introduced a logic-level quantum Boolean oracle synthesis method that combines resource-aware neural MCTS, affine Boolean preconditioning, and a resource-adaptive portfolio guard. On the current benchmark, the affine neural component substantially reduces T-count versus direct ANF, logical-AND direct ANF, fixed-coordinate MCTS, and SSHR-H on supported functions. On the $n \leq 6$ traditional-baseline slice, Pareto-Resource-NMCTS further improves the balanced weighted score over Resource-NMCTS, dominates internal ESOP cube-beam candidates by weighted score, and gives a clear same-benchmark improvement over both internal ESOP-MILP and exported ABC-ESOP baselines. Resource-profile and large_resource_core benchmarks show that this advantage is robust across several weighted objectives and larger random ANF instances. Exported SSHR, ABC-AIG, ABC-XAG, ABC-LUT, BDD, and ABC-ESOP baselines add independent cross-tool evidence that the gains are not an artifact of the internal experiment harness, while also showing that CNOT and depth remain important limitations. The new high-dimensional stress tests show that bounded guards scale without timeouts, beat FPRM-greedy on 60 of 64 weighted-score comparisons at $n = 14$ and 30 of 32 comparisons at $n = 15$, complete all 168 $n = 16$ ultra-scale rows while retaining score wins against FPRM root-beam, complete all 84 $n = 18$ mega-scale rows under the fast linear-pair guard, and beat exported ABC AIG/XAG/LUT and BDD on all 528 high-dimensional weighted-score comparisons. The next step is to reproduce ROS/mockturtle-style toolchain baselines and hardware-aware mappings so the empirical claim is submission-ready.

References

- [1] Robert K. Brayton and Alan Mishchenko. ABC: An academic industrial-strength verification tool. In *Computer Aided Verification*, volume 6174 of *Lecture Notes in Computer Science*, pages 24–40. Springer, 2010.
- [2] David Kremer, Ali Javadi-Abhari, and Priyanka Mukhopadhyay. Optimizing the non-Clifford-count in unitary synthesis using reinforcement learning, 2025. arXiv:2509.21709.
- [3] Mulundano Machiya, Matt Menickelly, Paul Hovland, and Ji Liu. MonteQ: A monte carlo tree search based quantum circuit synthesis framework, 2026. arXiv:2604.19029.
- [4] Giulia Meuli, Mathias Soeken, Earl Campbell, Martin Roetteler, and Giovanni De Micheli. The role of multiplicative complexity in compiling low T -count oracle circuits. In *2019 IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, pages 1–8. IEEE, 2019.
- [5] Giulia Meuli, Mathias Soeken, and Giovanni De Micheli. Xor-and-inverter graphs for quantum compilation. *npj Quantum Information*, 8(1):7, 2022.
- [6] Giulia Meuli, Mathias Soeken, Martin Roetteler, and Giovanni De Micheli. ROS: Resource-constrained oracle synthesis for quantum computers. *Electronic Proceedings in Theoretical Computer Science*, 318:119–130, 2020.
- [7] Sebastian Rietsch, Abhishek Y. Dubey, Christian Ufrecht, Maniraman Periyasamy, Axel Plinge, Christopher Mutschler, and Daniel D. Scherer. Unitary synthesis of Clifford+T circuits with reinforcement learning. In *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 824–835. IEEE, 2024.

- [8] Jordi Riu, Jan Nogu  , Gerard Vilaplana, Artur Garcia-Saez, and Marta P. Estarellas. Reinforcement learning based quantum circuit optimization via ZX-calculus. *Quantum*, 9:1758, 2025.
- [9] Dimitrios Tsaras, Antoine Grosnit, Lei Chen, Zhiyao Xie, Haitham Bou-Ammar, and Mingxuan Yuan. ShortCircuit: AlphaZero-driven circuit design, 2024. arXiv:2408.09858.
- [10] Robert Wille and Rolf Drechsler. BDD-based synthesis of reversible logic for large functions. In *Proceedings of the 46th Annual Design Automation Conference, DAC '09*, pages 270–275. ACM, 2009.
- [11] Mingfei Yu, Alessandro Tempia Calvino, Mathias Soeken, and Giovanni De Micheli. Back-end-aware fault-tolerant quantum oracle synthesis. In *2025 30th Asia and South Pacific Design Automation Conference (ASP-DAC)*, pages 930–937. ACM, 2025.
- [12] Qiang Zheng, Yongzhen Xu, Jiayi Zhang, Zhaofeng Su, and Shenggen Zheng. CNOT oriented synthesis for small-scale boolean functions using spatial structures of parallelotopes. In *2025 IEEE/ACM International Conference on Computer Aided Design (ICCAD)*, pages 1–9. IEEE, 2025.

Table 4: Resource-NMCTS compared with exported external baselines on the 177-function $n \leq 6$ extension. SSHR-I rows use an 8 s per-call Gurobi budget. ABC-AIG rows use verified ABC-optimized BLIF and Bennett-style resource estimates; ABC-XAG rows use verified ABC GIA/XOR-aware BLIF and logical XAG estimates; ABC-LUT rows use verified LUT-mapped BLIF and local-ANF node estimates; BDD rows use verified reduced ordered BDDs and Shannon-network estimates; ABC-ESOP rows use verified `&exorcism -q` ESOP-PLA and logical-AND cube estimates. Wins/losses/ties count common functions; negative mean relative values favor Resource-NMCTS.

External baseline	Metric	Wins	Losses	Ties	Mean relative
ABC-AIG export	T-count	173	2	2	-42.17%
ABC-AIG export	CNOT	177	0	0	-51.88%
ABC-AIG export	Depth	50	126	1	+52.86%
ABC-AIG export	Peak ancilla	177	0	0	-86.08%
ABC-AIG export	Score	177	0	0	-55.56%
ABC-XAG export	T-count	177	0	0	-53.44%
ABC-XAG export	CNOT	174	3	0	-37.28%
ABC-XAG export	Depth	30	144	3	+86.42%
ABC-XAG export	Peak ancilla	177	0	0	-89.70%
ABC-XAG export	Score	177	0	0	-64.15%
ABC-LUT export	T-count	175	0	2	-77.47%
ABC-LUT export	CNOT	177	0	0	-73.59%
ABC-LUT export	Depth	177	0	0	-71.97%
ABC-LUT export	Peak ancilla	176	0	1	-60.10%
ABC-LUT export	Score	177	0	0	-76.93%
BDD export	T-count	177	0	0	-65.70%
BDD export	CNOT	177	0	0	-61.37%
BDD export	Depth	177	0	0	-65.01%
BDD export	Peak ancilla	177	0	0	-82.11%
BDD export	Score	177	0	0	-68.02%
ABC-ESOP export	T-count	153	8	16	-24.26%
ABC-ESOP export	CNOT	95	67	15	-3.45%
ABC-ESOP export	Depth	146	23	8	-14.89%
ABC-ESOP export	Peak ancilla	67	2	108	-11.25%
ABC-ESOP export	Score	157	14	6	-21.61%
SSHR-H export	T-count	173	1	3	-45.61%
SSHR-H export	CNOT	41	131	5	+25.03%
SSHR-H export	Score	173	4	0	-38.87%
SSHR-I CNOT-opt	T-count	168	1	8	-36.96%
SSHR-I CNOT-opt	CNOT	0	168	9	+70.26%
SSHR-I CNOT-opt	Score	172	5	0	-29.48%
SSHR-I T-opt	T-count	168	1	8	-34.81%
SSHR-I T-opt	CNOT	4	161	12	+41.11%
SSHR-I T-opt	Score	169	7	1	-27.67%

Table 5: Exact bounded FPRM-DP comparison on the 72-function $n \leq 4$ traditional subset. The dynamic program is exact over the configured fixed-polarity FPRM factor model with monomial factors and CNOT-only linear-pair factors; it is not a global reversible-circuit optimum. Negative mean score changes favor the left-hand method in each comparison.

Comparison	Pairs	Score W/L/T	Mean Δ score
Resource-NMCTS vs Exact FPRM-DP	72	51/3/18	-12.18%
Pareto-Resource-NMCTS vs Exact FPRM-DP	72	51/0/21	-12.20%
Exact FPRM-DP vs ESOP MILP	72	57/12/3	-13.30%
Exact FPRM-DP vs SSHR-I CNOT	72	60/12/0	-13.73%
Exact FPRM-DP vs SSHR-I T	72	59/12/1	-13.38%

Table 6: Exact XAG multiplicative-complexity lower-bound comparison on the same 72-function $n \leq 4$ subset. The exact search minimizes the number of AND nodes in an XOR-AND graph; $4 \times \text{AND}$ is a global lower bound on logical-AND T-count, but not a CNOT or depth optimum.

Method	Pairs	On lower	Above lower	Mean T gap
Resource-NMCTS	72	12	60	+53.01%
Pareto-Resource-NMCTS	72	12	60	+53.01%
ESOP-MILP	72	3	69	+120.14%
SSHR-I-T	72	5	67	+143.06%

Table 7: Resource-NMCTS compared with exported ABC-AIG, ABC-XAG, ABC-LUT, and BDD baselines on the high-dimensional $n = 14$ and $n = 15$ random-ANF suites. Rows use verified ABC-optimized or LUT-mapped BLIF, or verified reduced ordered BDDs, with logic-level resource estimates. Negative mean relative values favor Resource-NMCTS.

Dataset	Metric	Functions	Wins	Losses	Ties	Mean relative
$n = 14$ ABC-AIG	T-count	64	64	0	0	-91.55%
$n = 14$ ABC-AIG	CNOT	64	64	0	0	-93.92%
$n = 14$ ABC-AIG	Depth	64	14	50	0	+1410.79%
$n = 14$ ABC-AIG	Peak ancilla	64	64	0	0	-99.79%
$n = 14$ ABC-AIG	Score	64	64	0	0	-94.13%
$n = 14$ ABC-XAG	T-count	64	64	0	0	-93.61%
$n = 14$ ABC-XAG	CNOT	64	64	0	0	-92.42%
$n = 14$ ABC-XAG	Depth	64	14	50	0	+1160.89%
$n = 14$ ABC-XAG	Peak ancilla	64	64	0	0	-99.84%
$n = 14$ ABC-XAG	Score	64	64	0	0	-95.48%
$n = 14$ ABC-LUT	T-count	64	64	0	0	-97.30%
$n = 14$ ABC-LUT	CNOT	64	64	0	0	-97.20%
$n = 14$ ABC-LUT	Depth	64	64	0	0	-97.15%
$n = 14$ ABC-LUT	Peak ancilla	64	64	0	0	-99.46%
$n = 14$ ABC-LUT	Score	64	64	0	0	-97.44%
$n = 14$ BDD	T-count	64	64	0	0	-92.45%
$n = 14$ BDD	CNOT	64	64	0	0	-92.39%
$n = 14$ BDD	Depth	64	64	0	0	-93.25%
$n = 14$ BDD	Peak ancilla	64	64	0	0	-99.50%
$n = 14$ BDD	Score	64	64	0	0	-93.24%
$n = 15$ ABC-AIG	T-count	32	32	0	0	-92.28%
$n = 15$ ABC-AIG	CNOT	32	32	0	0	-94.40%
$n = 15$ ABC-AIG	Depth	32	7	25	0	+1943.38%
$n = 15$ ABC-AIG	Peak ancilla	32	32	0	0	-99.67%
$n = 15$ ABC-AIG	Score	32	32	0	0	-94.59%
$n = 15$ ABC-XAG	T-count	32	32	0	0	-94.87%
$n = 15$ ABC-XAG	CNOT	32	32	0	0	-93.89%
$n = 15$ ABC-XAG	Depth	32	6	26	0	+1408.93%
$n = 15$ ABC-XAG	Peak ancilla	32	32	0	0	-99.81%
$n = 15$ ABC-XAG	Score	32	32	0	0	-96.33%
$n = 15$ ABC-LUT	T-count	32	32	0	0	-97.64%
$n = 15$ ABC-LUT	CNOT	32	32	0	0	-97.53%
$n = 15$ ABC-LUT	Depth	32	32	0	0	-97.42%
$n = 15$ ABC-LUT	Peak ancilla	32	32	0	0	-99.40%
$n = 15$ ABC-LUT	Score	32	32	0	0	-97.76%
$n = 15$ BDD	T-count	32	32	0	0	-94.14%
$n = 15$ BDD	CNOT	32	32	0	0	-94.14%
$n = 15$ BDD	Depth	32	32	0	0	-94.77%
$n = 15$ BDD	Peak ancilla	32	32	0	0	-99.65%
$n = 15$ BDD	Score	32	32	0	0	-94.75%

Table 8: Resource-NMCTS compared with exported ABC-AIG, ABC-XAG, ABC-LUT, and BDD baselines on the ultra_highdim_resource $n = 16$ scale check. Score columns report weighted-score wins/losses/ties and mean relative score. Depth columns expose the ABC depth counterweight.

Baseline	Score W/L/T	Depth W/L/T	T rel.	CNOT rel.	Depth rel.	Score rel.
ABC-AIG	24/0/0	2/22/0	−96.05%	−97.24%	+2171.79%	−97.29%
ABC-XAG	24/0/0	2/22/0	−96.98%	−96.52%	+1453.24%	−97.88%
ABC-LUT	24/0/0	24/0/0	−98.94%	−98.95%	−98.97%	−99.00%
BDD	24/0/0	24/0/0	−96.42%	−96.45%	−96.90%	−96.81%

Table 9: Resource-NMCTS compared with exported ABC-AIG, ABC-XAG, ABC-LUT, and BDD baselines on the mega_highdim_resource $n = 18$ stress check. Score columns report weighted-score wins/losses/ties and mean relative score. Depth columns expose the ABC depth counterweight.

Baseline	Score W/L/T	Depth W/L/T	T rel.	CNOT rel.	Depth rel.	Score rel.
ABC-AIG	12/0/0	2/10/0	−98.19%	−98.74%	+3429.31%	−98.76%
ABC-XAG	12/0/0	1/11/0	−98.55%	−98.33%	+1586.42%	−98.98%
ABC-LUT	12/0/0	12/0/0	−99.65%	−99.64%	−99.64%	−99.66%
BDD	12/0/0	12/0/0	−97.97%	−97.99%	−98.24%	−98.19%

Table 10: Ablation results on the same 322-function suite. Wins, ties, and losses use the composite resource score unless otherwise stated.

Comparison	Wins	Ties	Losses	Mean relative score
Affine-NMCTS vs affine-greedy	153	169	0	−2.82%
affine-no-guard vs affine-greedy	65	257	0	−1.08%
Affine-NMCTS vs affine-no-guard	88	234	0	−1.74%
Affine-NMCTS vs fixed-coordinate MCTS	165	156	0	−14.82%

Table 11: Learned-prior ablation on the $n \leq 6$ traditional_resource slice. Rows compare the learned action scorer against the same search with only the heuristic action prior. Negative mean score change favors the learned prior.

Method	Score wins	Score losses	Score ties	Mean score change
Affine-NMCTS	42	0	135	−1.47%
Resource-NMCTS	39	0	138	−1.10%
Pareto-Resource-NMCTS	29	0	148	−0.78%

Table 12: High-dimensional root-action teacher diagnostic on 10 $n = 14$ functions with available linear-pair actions. Oracle rows score wider root-action sets by building actual greedy child plans. Negative mean score change favors the target.

Target	Baseline	Score W/L/T	Mean Δ score
oracle top-12	heuristic top-4	3/0/7	−0.18%
oracle top-24	heuristic top-4	3/0/7	−0.18%
oracle top-24	heuristic top-1	7/0/3	−0.43%
neural top-4	heuristic top-4	1/1/8	+0.06%

Table 13: Contribution decomposition from matched function-level comparisons. Wins/losses/ties use the composite resource score. Negative mean score change favors the component in the first column.

Component	Dataset	Pairs	Score W/L/T	Mean Δ score
Affine greedy vs fixed-coordinate MCTS	ablation_affine	321	165/88/68	-12.12%
Neural refine over affine greedy	ablation_affine	322	65/0/257	-1.08%
Guarded Affine-NMCTS over no-guard	ablation_affine	322	88/0/234	-1.74%
Learned prior for Resource-NMCTS	traditional_resource	177	39/0/138	-1.10%
Pareto archive over Resource-NMCTS	traditional_resource	177	68/0/109	-3.26%
No-MCTS portfolio over heuristic-only	trad. ablation	177	54/0/123	-2.51%
Resource-NMCTS over no-MCTS portfolio	trad. ablation	177	54/0/123	-1.44%
Pareto Resource-NMCTS over no-MCTS portfolio	trad. ablation	177	106/0/71	-4.69%
Highdim no-MCTS portfolio over root beam	n14 ablation	16	14/0/2	-6.25%
Highdim no-MCTS portfolio over linear-pair	n14 ablation	16	14/0/2	-3.08%
Linear-pair guard vs root beam at n=14	highdim_resource	64	60/0/4	-3.00%
Recursive linear-pair guard vs root beam at n=15	n15 scale	32	30/0/2	-5.28%
Linear-pair guard vs root beam at n=16	n16 scale	24	22/0/2	-1.88%
Fast linear-pair guard vs root beam at n=18	n18 stress	12	6/0/6	-1.91%
Resource-NMCTS vs fast linear-pair at n=18	n18 stress	12	12/0/0	-3.55%

Table 14: Runtime distribution on the ablation_affine suite. SSHR-H is skipped for $n > 6$ by the harness.

Method	Completed	Errors	Skipped	Median s	p95 s	Max s
Direct ANF	322	0	0	0.00	0.00	0.01
AND-direct ANF	322	0	0	0.00	0.00	0.01
Fixed MCTS	321	1	0	0.08	12.26	89.90
Affine-greedy	322	0	0	0.03	0.23	1.82
Affine-no-guard	322	0	0	0.09	3.62	4.85
Affine-NMCTS	322	0	0	0.61	13.67	300.02
SSHR-H	177	0	145	0.00	0.12	0.70

Table 15: Mean logical resources on completed rows in the ablation_affine suite. SSHR-H means are over its supported $n \leq 6$ subset and therefore are not directly comparable to all-suite means.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	491.9	275.2	275.4	1.28	509.8
AND-direct ANF	254.1	405.8	406.1	2.11	282.1
Fixed MCTS	155.2	291.4	291.7	1.85	176.1
Affine-greedy	154.1	280.8	283.6	1.99	174.7
Affine-no-guard	153.8	280.3	282.9	1.93	174.2
Affine-NMCTS	145.4	272.7	275.4	1.89	165.3
SSHR-H	81.0	67.6	88.7	1.36	88.2

Table 16: Objective-score best-or-tied counts in the resource-profile sweep. Rows may sum to more than 47 because several methods can achieve the same minimum score on a function.

Profile	Pareto	Profile	Pol.	Res.	Aff.	MCTS	SSHR	Direct	ESOP
T-heavy	44	25	24	25	25	6	3	5	1
Balanced	44	25	24	25	25	6	3	5	1
CNOT-depth	42	22	26	22	22	6	5	5	1
Ancilla-tight	43	29	19	28	27	7	4	5	1

Table 17: Mean logical resources on completed rows in the large_resource_core benchmark. The fixed-MCTS mean excludes five timeout rows, so its completed-row average is censored toward easier instances.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	809.3	422.8	423.1	1.28	835.5
AND-direct ANF	413.3	655.4	655.7	2.14	456.1
Fixed MCTS	215.2	399.3	399.6	1.89	242.5
Affine-NMCTS	245.3	448.9	451.0	1.95	275.7
Resource-NMCTS	243.4	447.0	449.5	1.95	273.7
Profile-Resource-NMCTS	255.6	457.6	460.0	1.96	286.5

Table 18: Runtime distribution on the large_resource_core benchmark.

Method	Completed	Errors	Skipped	Median s	p95 s	Max s
Direct ANF	330	0	0	0.00	0.01	0.03
AND-direct ANF	330	0	0	0.00	0.01	0.05
Fixed MCTS	325	5	0	0.10	71.54	273.95
Affine-NMCTS	330	0	0	0.72	120.45	300.42
Resource-NMCTS	330	0	0	1.31	58.86	300.85
Profile-Resource-NMCTS	330	0	0	2.29	25.68	67.61

Table 19: Mean logical resources on the highdim_resource $n = 14$ stress benchmark.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	7159.4	3381.1	3381.2	1.22	7350.2
AND-direct ANF	3597.6	5642.0	5642.2	2.03	3932.4
FPRM-greedy	2263.7	3894.8	3894.9	2.03	2497.0
FPRM root beam	2248.8	3886.6	3886.8	2.03	2481.6
FPRM linear pair	2193.5	3790.8	3791.8	2.94	2422.5
Affine-greedy	2263.7	3894.8	3894.9	2.03	2497.0
Resource-NMCTS	2149.5	3724.9	3726.8	3.06	2374.9
Profile-Resource-NMCTS	2149.5	3724.9	3726.8	3.06	2374.9
Pareto-Resource-NMCTS	2148.8	3725.0	3726.7	3.06	2374.2

Table 20: Runtime distribution on the highdim_resource $n = 14$ stress benchmark.

Method	Completed	Errors	Skipped	Median s	p95 s	Max s
Direct ANF	64	0	0	0.02	0.06	0.08
AND-direct ANF	64	0	0	0.02	0.04	0.07
FPRM-greedy	64	0	0	3.46	65.99	85.56
FPRM root beam	64	0	0	2.83	37.23	49.27
FPRM linear pair	64	0	0	2.57	30.76	41.59
Affine-greedy	64	0	0	6.65	107.89	129.18
Resource-NMCTS	64	0	0	8.57	80.43	121.22
Profile-Resource-NMCTS	64	0	0	9.29	84.23	122.85
Pareto-Resource-NMCTS	64	0	0	67.79	300.02	300.41

Table 21: Mean logical resources on the highdim_scale_resource $n = 15$ scaling check.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	10591.8	4964.2	4964.3	1.16	10870.7
AND-direct ANF	5320.0	8339.9	8339.9	1.94	5812.8
FPRM-greedy	3315.6	5681.3	5681.3	1.94	3653.8
FPRM root beam	3302.5	5670.2	5670.2	1.94	3640.1
FPRM linear pair	3230.0	5545.5	5546.1	2.84	3562.0
FPRM linear pair deep	3200.2	5492.4	5493.6	3.06	3529.6
FPRM linear parity	3229.9	5545.3	5545.9	2.84	3561.9
Resource-NMCTS	3200.2	5492.4	5493.6	3.06	3529.6
Profile-Resource-NMCTS	3200.2	5492.4	5493.6	3.06	3529.6
Pareto-Resource-NMCTS	3199.6	5492.8	5494.0	3.06	3529.0

Table 22: Runtime distribution on the highdim_scale_resource $n = 15$ scaling check.

Method	Completed	Errors	Skipped	Median s	p95 s	Max s
Direct ANF	32	0	0	0.03	0.05	0.05
AND-direct ANF	32	0	0	0.03	0.08	0.11
FPRM-greedy	32	0	0	1.47	44.43	46.58
FPRM root beam	32	0	0	3.03	59.29	70.22
FPRM linear pair	32	0	0	3.94	81.93	95.82
FPRM linear pair deep	32	0	0	10.30	86.72	95.55
FPRM linear parity	32	0	0	4.10	78.67	94.47
Resource-NMCTS	32	0	0	9.70	90.02	94.55
Profile-Resource-NMCTS	32	0	0	11.33	93.68	102.74
Pareto-Resource-NMCTS	32	0	0	117.52	300.30	300.64

Table 23: Mean logical resources on the ultra_highdim_resource $n = 16$ scale check.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	12053.7	5616.0	5616.1	1.38	12369.5
AND-direct ANF	6047.7	9491.2	9491.3	2.29	6608.7
FPRM root beam	3822.3	6613.5	6613.6	2.29	4216.2
FPRM linear pair	3759.7	6501.3	6501.5	3.21	4148.8
Resource-NMCTS	3759.7	6501.3	6501.5	3.21	4148.8
Profile-Resource-NMCTS	3759.7	6501.3	6501.5	3.21	4148.8
Pareto-Resource-NMCTS	3756.3	6504.2	6504.3	3.21	4145.6

Table 24: Runtime distribution on the ultra_highdim_resource $n = 16$ scale check.

Method	Completed	Errors	Skipped	Median s	p95 s	Max s
Direct ANF	24	0	0	0.06	0.06	0.06
AND-direct ANF	24	0	0	0.06	0.06	0.06
FPRM root beam	24	0	0	5.93	21.19	31.91
FPRM linear pair	24	0	0	8.03	30.18	42.64
Resource-NMCTS	24	0	0	7.88	27.83	41.72
Profile-Resource-NMCTS	24	0	0	8.43	29.77	44.92
Pareto-Resource-NMCTS	24	0	0	130.92	301.10	303.56

Table 25: Mean logical resources on the mega_highdim_resource $n = 18$ stress check.

Method	Mean T	Mean CNOT	Mean depth	Mean ancilla	Mean score
Direct ANF	21460.7	9794.7	9794.8	1.25	22008.9
AND-direct ANF	10756.0	16840.5	16840.6	2.17	11747.4
FPRM root beam	6765.3	11585.2	11585.2	2.17	7451.3
FPRM linear-pair fast	6754.0	11566.8	11568.4	2.67	7439.9
Resource-NMCTS	6641.7	11382.4	11383.7	3.25	7317.9
Profile-Resource-NMCTS	6641.7	11382.4	11383.7	3.25	7317.9
Pareto-Resource-NMCTS	6641.7	11382.4	11383.7	3.25	7317.9

Table 26: Runtime distribution on the mega_highdim_resource $n = 18$ stress check.

Method	Completed	Errors	Skipped	Median s	p95 s	Max s
Direct ANF	12	0	0	0.72	0.73	0.74
AND-direct ANF	12	0	0	0.72	0.73	0.73
FPRM root beam	12	0	0	65.11	141.05	162.16
FPRM linear-pair fast	12	0	0	65.48	138.67	160.92
Resource-NMCTS	12	0	0	83.31	173.18	205.10
Profile-Resource-NMCTS	12	0	0	83.34	173.78	205.33
Pareto-Resource-NMCTS	12	0	0	83.50	172.93	205.44